

MA.912.F.1.5

Benchmark

MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables or written descriptions.

Benchmark Clarifications:

Clarification 1: Key features are limited to domain; range; intercepts; slope and end behavior.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.2.4
- MA.912.AR.2.5

Terms from the K-12 Glossary

- Domain
- Intercept
- Range
- Slope

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.5

Next Benchmarks

- MA.912.F.1.7

Purpose and Instructional Strategies

In grade 8, students interpreted the slope and y -intercept of a linear equation in two variables. In Algebra I, students compare key features of two or more linear functions. In later courses, students will compare key features of linear and nonlinear functions.

- Instruction includes the use of various forms of linear equations. Additionally, linear functions can be represented as a table of values or graphically.
 - Standard Form
Can be described by the equation $Ax + By = C$, where A , B and C are any rational number.
 - Slope-Intercept Form
Can be described by the equation $y = mx + b$, where m is the slope and b is the y -intercept.
 - Point-Slope Form
Can be described by the equation $y - y_1 = m(x - x_1)$, where (x_1, y_1) are a point on the line and m is the slope of the line.
- Problem types include comparing linear functions presented in similar forms and in different forms, and comparing more than two linear functions.
- Instruction includes representing domain and range using words, inequality notation and set-builder notation.
 - Words
If the domain is all real numbers, it can be written as “all real numbers” or “any value of x , such that x is a real number.”
 - Inequality notation
If the domain is all values of x greater than 2, it can be represented as $x > 2$.
 - Set-builder notation
If the domain is all values of x less than or equal to zero, it can be represented as $\{x|x \leq 0\}$ and is read as “all values of x such that x is less than or equal to zero.”

Common Misconceptions or Errors

- When describing domain or range, students may assign their constraints to the incorrect variable. In these cases, ask reflective questions to help students examine the meaning of the domain and range in the problem.
- Students may also miss the need for compound inequalities when describing domain or range. In these cases, use a graph of the function to point out areas of their constraint that would not make sense in context.

Strategies to Support Tiered Instruction

- Teacher provides a laminated cue card to aid in the identification of domain and range restrictions:
 - Is the constraint on the independent or dependent variable in the context of the problem?
 - Does the constraint restrict the input or output value in the context of the problem?
 - Was the constraint shown or highlighted on the x - or y -axis?
- Teacher provides a chart to show different terminology associated with domain and range.

<i>Domain</i>	<i>Range</i>
Independent Variable	Dependent Variable
x -values	y -values
Input	Output

- Instruction includes opportunities to use a graphic organizer to chart and provide specific examples of domain and range as compound inequalities.

Compound Inequality	Example(s)	Notation
AND	Between 3 pm and 6 pm	$3 \leq x \leq 6$
OR	Under 10 years or at least 70 years	$x < 10$ or $x \geq 70$

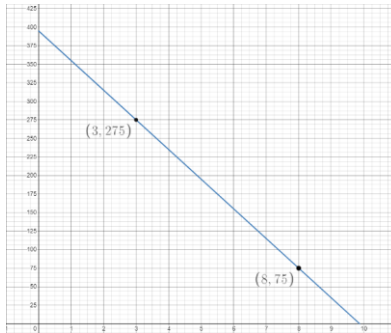
- Teacher provides a graphic organizer for each of the three forms of linear equations (standard, slope-intercept and point-slope form) that can be co-created to highlight key similarities and differences.

Teacher and student write the equation that shows this form chosen.	Teacher can ask student to describe different features of the selected form or ways this form is different from other forms. Teacher asks student what are some advantages or disadvantages of the selected form.
Teacher and student select one of the three forms and place name here.	
Teacher and student show several examples. Be sure to show examples where a coefficient of x or y is zero, negative or one.	Teacher and student write linear equations in other forms, or in none of the forms.

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.5.1)

Callie and her friend Elena are reading through different novels they checked out from their school library last Tuesday. Callie's progress through her novel can be modeled by the function $p(d) = -25d + 318$, where $p(d)$ represents the number of pages remaining to be read and d is the number of days since receiving the book. Elena's progress through her novel is modeled by the graph below.



Part A. Which student's novel has more pages to read?

Part B. Assuming they both continue to read at a constant rate, which student will complete their novel first?

Instructional Items

Instructional Item 1

Two linear functions, $f(x)$ and $g(x)$, are represented below. Compare the functions by stating which has a greater y-intercept, x-intercept and rate of change.

$$f(x) = 4x - 3$$

x	-3	-1	1	3
$g(x)$	13	8	3	-2

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*